

LAHORE CITY, Pakistan

WMO: 416400

Lat: 31.550N

Lon: 74.333E

Elev: 705

StdP: 14.33

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.3	46.2	32.3	27.5	63.3	35.2	30.9	61.9	10.6	61.9	8.9	61.7	0.3	310	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	17.3	108.1	74.2	105.6	73.7	102.9	74.2	84.2	93.4	83.4	92.7	82.7	92.1	3.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.2	172.5	89.5	81.3	166.9	89.1	80.5	162.6	88.8	49.4	92.7	48.5	93.6	47.7	92.6	93.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.3	7.6	DB	40.4	112.1	2.9	2.3	38.4	113.7	36.7	115.1	35.1	116.4	33.0	118.0	(4)
(5)			WB	38.1	86.2	2.7	1.2	36.1	87.0	34.6	87.7	33.0	88.4	31.1	89.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	77.4	56.2	62.7	72.2	83.3	91.0	92.2	89.0	87.7	85.9	79.8	69.1	59.2	(6)
(7)		DBStd	13.29	4.06	4.94	6.19	5.89	5.20	5.71	4.08	3.53	3.57	4.46	3.69	4.40	(7)
(8)		HDD50	7	5	0	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD65	572	274	96	12	0	0	0	0	0	0	0	6	184	(9)
(10)		CDD50	10014	197	356	689	999	1271	1266	1208	1169	1076	923	573	287	(10)
(11)		CDD65	5104	0	32	236	549	806	816	743	704	626	458	130	4	(11)
(12)		CDH74	70566	2	146	2028	7204	12829	13306	11162	10025	8391	4726	723	24	(12)
(13)		CDH80	41502	0	7	717	4044	8584	9109	6792	5677	4412	2068	92	0	(13)
(14)	Wind	WSAvg	1.9	1.2	2.0	2.5	3.0	3.0	2.8	2.3	1.8	1.5	1.2	0.9	0.8	(14)
(15)	Precipitation	PrecAvg	24.3	0.9	1.1	1.5	0.7	0.8	2.1	7.4	6.1	2.8	0.4	0.2	0.4	(15)
(16)		PrecMax	44.0	4.8	4.6	6.6	5.6	4.4	8.2	18.8	21.9	17.7	4.2	2.5	4.4	(16)
(17)		PrecMin	11.8	0.0	0.0	0.0	0.0	0.0	0.0	1.2	0.1	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	8.2	1.0	1.1	1.3	0.9	0.9	2.0	4.0	4.4	2.7	0.6	0.4	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.7	81.0	94.4	105.7	111.0	112.1	104.7	99.2	98.0	95.3	85.4	76.7	(19)
(20)			MCWB	58.4	63.7	70.2	70.0	72.4	74.5	80.1	80.8	78.1	75.3	65.9	59.8	(20)
(21)		2%	DB	69.6	78.0	90.7	102.3	108.0	109.0	101.0	97.4	96.1	93.4	82.5	73.9	(21)
(22)			MCWB	56.3	61.7	68.6	69.6	72.6	75.4	81.0	81.1	78.3	73.9	65.5	59.1	(22)
(23)		5%	DB	67.5	75.4	87.1	99.1	105.7	106.3	98.8	95.9	94.7	91.7	80.5	71.3	(23)
(24)			MCWB	55.2	60.4	67.6	68.9	72.2	75.6	81.6	81.2	78.4	73.2	64.5	57.6	(24)
(25)		10%	DB	65.3	72.6	83.9	95.6	103.1	103.5	96.7	94.3	93.2	89.6	78.4	68.8	(25)
(26)			MCWB	54.1	59.3	65.9	68.4	72.0	76.4	81.8	81.3	78.1	72.1	63.6	56.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.2	66.2	73.3	74.3	79.3	84.3	87.0	84.6	83.4	79.9	70.2	63.3	(27)
(28)			MCD B	68.9	76.6	88.5	96.2	99.8	97.9	94.0	92.3	91.8	90.3	80.0	71.9	(28)
(29)		2%	WB	59.2	64.3	71.2	72.4	77.3	82.8	84.6	83.7	82.2	78.5	68.4	61.4	(29)
(30)			MCD B	66.4	74.5	85.7	94.0	98.7	96.1	93.5	91.7	91.0	88.3	78.1	70.4	(30)
(31)		5%	WB	57.5	62.8	69.6	71.1	75.9	81.7	83.7	83.1	81.4	76.5	67.0	59.5	(31)
(32)			MCD B	63.8	72.8	83.5	92.4	97.7	95.2	93.1	91.2	90.2	86.8	76.6	68.6	(32)
(33)		10%	WB	55.8	61.1	67.8	70.0	74.6	80.7	83.0	82.5	80.6	74.6	65.5	57.7	(33)
(34)			MCD B	62.8	70.0	81.1	91.5	96.6	94.7	92.6	90.8	89.4	86.4	75.3	66.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.2	16.4	17.6	20.6	20.7	17.3	12.1	11.1	13.3	17.7	19.1	17.9	(35)
(36)			MCDBR	20.0	18.9	20.7	24.3	23.1	21.3	14.4	12.7	14.4	18.1	20.8	21.5	(36)
(37)			MCWBR	9.6	8.1	6.8	6.7	6.9	6.2	4.8	4.4	4.7	5.1	7.7	9.9	(37)
(38)		5% WB	MCDBR	16.4	16.8	18.9	21.4	20.3	15.0	12.5	11.3	12.3	15.2	17.3	18.7	(38)
(39)			MCWBR	9.0	8.0	7.3	7.4	7.9	6.4	5.7	4.8	5.1	5.6	7.8	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.488	0.495	0.513	0.574	0.653	0.748	0.790	0.699	0.608	0.659	0.623	0.537	(40)	
(41)		taud	1.855	1.889	1.879	1.788	1.653	1.527	1.498	1.653	1.784	1.596	1.614	1.755	(41)	
(42)		Ebn at Noon	220	237	245	235	218	198	188	204	217	188	179	195	(42)	
(43)		Edh at Noon	53	56	61	69	80	91	93	79	67	75	68	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	907	1246	1664	1991	2200	2050	1743	1679	1650	1457	1118	934	(44)	
(45)		RadStd	105	128	106	89	76	118	110	128	88	71	62	56	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page